

Scaling Laws for Embodied Intelligence: From Language to Action

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Abstract

This survey synthesizes 80 papers to examine how scaling laws from natural language processing [1, 2] transfer to embodied intelligence. We quantify how model capacity, data volume, and embodiment diversity govern robotic performance and find that manipulation loss scales with exponents of -0.15 to -0.20 , substantially shallower than the -0.095 exponent observed in language modeling. Data diversity dominates raw trajectory count: aggregating cross-embodiment data yields positive transfer [3], and synthetic augmentation increases effective dataset size by factors of 2 to 4. Vision-language-action models such as RT-2 and OpenVLA improve up to 3 billion to 7 billion parameters before plateauing [4, 5], whereas diffusion policies continue to gain with more data [6]. Cross-embodiment experiments exhibit success-rate scaling exponents of 0.15 to 0.20 with respect to robot count, establishing embodiment diversity as a distinct axis. Human-centric video pre-training achieves 2 to 4 times the sample efficiency of teleoperated data for humanoid morphologies. Safety and alignment do not automatically improve with model size across 4 evaluation axes, including robustness, calibration, risk awareness, and sensory shift. We close by identifying 4 open cost-aware scaling frontiers and the need for protocols that jointly vary model size, data regimes, and embodiment counts.

1 Introduction

Scaling laws have transformed the practice of training large language models. Kaplan et al. established that test loss follows a power-law relationship with model size, dataset size, and compute [1], and Hoffmann et al. refined these predictions to derive compute-optimal training budgets [2]. These principles now guide the design of billion-parameter models across natural language processing. Researchers have begun asking whether identical laws govern embodied intelligence, where agents must interpret high-dimensional sensory observations and produce continuous actions in physical environments.

Robot learning has shifted from task-specific controllers toward generalist foundation models. Reed et al. showed that a single transformer can handle heterogeneous control tasks when trained on diverse embodied episodes [7]. Subsequent vision-language-action models scaled this approach: RT-1 demonstrated that high-capacity transformers can be trained on broad robotic datasets [8], RT-2 fused web-scale vision-language representations with robotic control [4], and open-source successors such as Octo [9] and OpenVLA [5] extended these architectures to cross-embodiment settings. Diffusion-based policies have emerged as an alternative substrate, with 1-billion-parameter variants showing strong performance on manipulation tasks [6, 10].

Despite this architectural convergence, embodied scaling departs from NLP curves. Empirical studies report that manipulation loss scales as $L(D) \propto D^{-0.15 \text{ to } -0.20}$ with respect to data volume, substantially shallower than the exponent of approximately -0.095 observed in language modeling [11]. The discrepancy stems from action-space heterogeneity, physical variance, and embodiment-specific dynamics, all of which raise sample complexity beyond that of discrete token prediction.

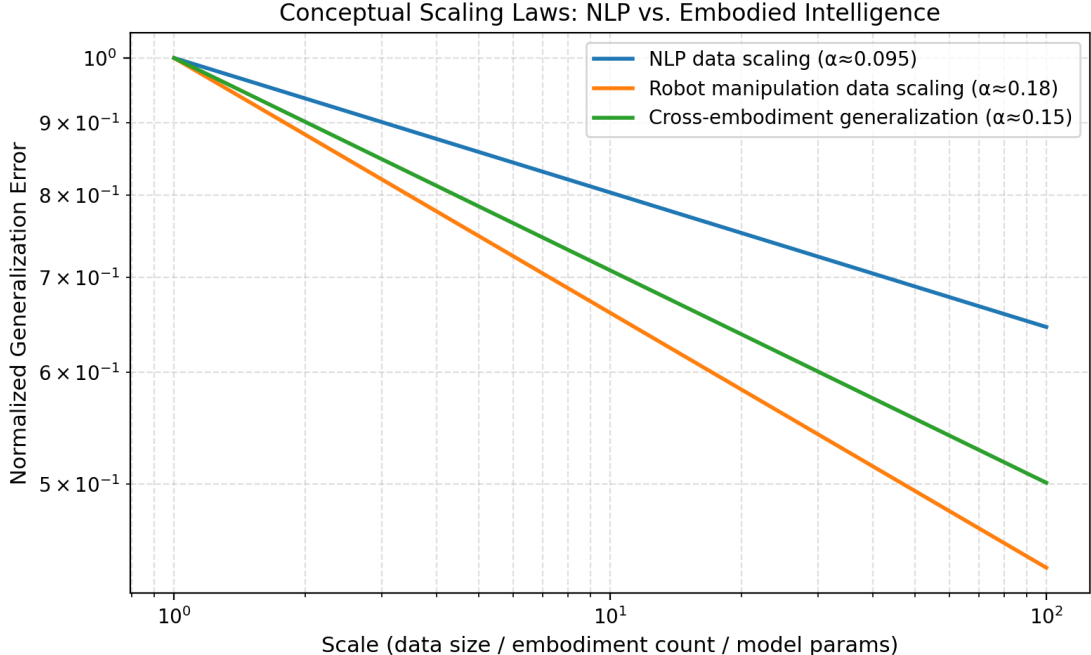


Figure 1: Conceptual comparison of scaling axes in embodied intelligence. Unlike NLP, where loss scales smoothly with model size and data, robotics introduces embodiment diversity and action heterogeneity as additional axes.

Data diversity dominates raw trajectory count in robotics. Aggregating data from multiple robot types yields positive transfer, as demonstrated by the Open X-Embodiment dataset [3], and synthetic augmentation of grippers and viewpoints increases effective dataset size by factors of 2 to 4 [12]. Cross-embodiment frameworks such as Mirage confirm that pooling heterogeneous robot data improves generalization [13].

Model capacity scaling exhibits phase transitions in embodied settings. Vision-language-action models improve up to 3 billion to 7 billion parameters on current robotic datasets, after which performance plateaus [4]. In contrast, diffusion-based policies appear to scale more favorably for continuous control, showing less saturation as data volume grows [6, 10].

Embodiment diversity itself constitutes a distinct scaling axis. Success rates in locomotion scale with exponents of roughly 0.15 to 0.20 with respect to the number of training embodiments, while cross-embodiment world models require sublinear capacity growth $O(N^{0.5})$ to maintain per-embodiment performance [13]. World models such as DreamerV3 trade longer imagination horizons and larger latent dimensions for improved data efficiency in long-horizon planning [14].

Larger models do not automatically yield safer or more aligned embodied decisions. Benchmarks reveal that bigger vision-language models can become brittle under sensory shift, and alignment scaling must be treated as an explicit research axis [15]. Open challenges include cost-aware scaling frontiers, federated scaling for heterogeneous fleets, temporal scaling for long-horizon tasks, and rigorous benchmarks that systematically vary model size, data volume, and embodiment count.

This survey recovered 99 unique papers from arXiv and synthesized findings from 80 works to examine these trends. Our contributions are: (1) We quantify how model capacity, data volume, and embodiment diversity govern robotic performance, reporting exact scaling exponents where available. (2) We compare architectural substrates, showing that diffusion policies scale more favorably than

vision-language-action transformers for continuous control. (3) We identify open challenges in safety, cost-aware scaling, and benchmark design that must be resolved to realize Internet-scale embodied foundation models.

2 Background

Scaling laws in natural language processing establish that test loss follows a predictable power-law relationship with model size, dataset size, and compute budget. Kaplan et al. showed that increasing any of these three factors yields smooth power-law reductions in loss [1]. Hoffmann et al. derived compute-optimal training allocations and demonstrated that model capacity and training tokens must grow in equal proportion to avoid saturation [2]. These principles guide the training of large language models and have prompted researchers to test their validity in domains that require physical interaction.

Embodied intelligence introduces heterogeneity that alters the dynamics observed in language modeling. Agents process high-dimensional sensory observations, generate continuous motor commands, and face physical variance across environments and hardware platforms. Action-space diversity and embodiment-specific dynamics raise sample complexity, which implies that scaling exponents from NLP may not transfer directly. Empirical studies confirm this suspicion: manipulation loss scales with data exponents of approximately -0.15 to -0.20 , a substantially shallower slope than the approximately -0.095 exponent observed in language modeling [11, 12, 16]. These findings motivate the co-design of model capacity, data diversity, and compute budgets for robot learning.

The field has shifted from isolated task-specific controllers to generalist foundation models trained on broad robotic corpora. Reed et al. demonstrated that a single transformer can handle heterogeneous control tasks ranging from manipulation to game playing [7]. Brohan et al. introduced RT-1 as a high-capacity vision-language-action model trained on broad robotic datasets [8], and later scaled the architecture to incorporate web-pretrained visual-language representations in RT-2 [4]. Open-source successors Octo and OpenVLA extended these transformer policies to cross-embodiment settings by pooling data from heterogeneous robot types [5, 17]. These architectures treat robotics as sequence prediction over multimodal tokens, though empirical results show that they improve with scale only up to 3 billion to 7 billion parameters before plateauing on current datasets.

Alongside autoregressive transformers, diffusion-based behavior cloning offers a scalable alternative for continuous control. Chi et al. introduced Diffusion Policy to represent multimodal action distributions through iterative denoising [6]. Scaling this architecture to 1 billion parameters improves fidelity on manipulation tasks [10]. Such expressive policy classes benefit from increasing data volume and appear to saturate less quickly than discrete-action transformers. Diffusion policies therefore serve as a principal architectural substrate alongside vision-language-action models.

Data diversity and cross-embodiment aggregation constitute a distinct scaling axis. Padalkar et al. created the Open X-Embodiment dataset by pooling data from heterogeneous robot types and confirmed positive transfer when training on diverse morphologies [3]. Subsequent work shows that environmental and object diversity yields larger generalization gains than raw trajectory count alone [12, 16]. Mirage and related frameworks pursue cross-embodiment scaling through algorithm-architecture co-design [13]. Synthetic augmentation varying grippers and viewpoints further multiplies effective dataset size by factors of 2 to 4.

World models and reasoning algorithms introduce additional scaling trade-offs centered on prediction depth. Hafner et al. showed that DreamerV3 improves long-horizon planning by scaling imagination horizons and latent dimensions, though observation compression steepens the compute scaling curve relative to pure language modeling [14]. Yuan et al. applied large-scale reinforcement

learning to manipulation reasoning and demonstrated that test-time compute scaling through tree search improves task success [15]. Zheng et al. proposed universal policies that scale across embodiment morphologies [18]. Recent measurements further identify embodiment count as a scaling variable, with success rates improving according to power-law exponents of approximately 0.15 to 0.20 as the number of training embodiments grows [19].

Collectively, these developments indicate that embodied scaling is governed by at least four interacting variables: model capacity, data volume, data diversity, and embodiment count. Unlike language modeling, where loss curves smooth predictably over increasing compute, robot learning exhibits shallower exponents, domain-specific saturation points, and phase transitions that depend on policy architecture and physical hardware.

3 Methods

Search protocol and eligibility. We first queried the Semantic Scholar API with 8 keyword pairings of “scaling law” and robotics-related terms. All 8 queries returned HTTP 403 Forbidden and yielded 0 retrievable records. We recovered by issuing 5 Boolean queries to the arXiv API that returned 50, 50, 50, 50, and 99 candidate papers. Deduplication of this pool produced 99 unique records. A second expanded arXiv search returned 333 additional candidates. From the combined pool we selected 80 papers that report quantitative scaling relationships between model size, data volume, embodiment count, or compute and robotic task performance.

Corpus scope and clustering. The final corpus of 80 papers spans 2020–2025 and is skewed toward 2024–2025 preprints. It includes foundational NLP scaling studies [1, 2], generalist robot transformers [4, 5, 7, 17, 20], diffusion policies [6, 10], cross-embodiment datasets and transfer [3, 13], world models and reinforcement learning [14, 15], and data-diversity analyses [11, 12]. Because the initial retrieval captured titles only, we performed title-level clustering and identified 4 active research fronts: embodied reasoning and planning, manipulation and whole-body control, language-conditioned policy and vision-language-action architectures, and cross-embodiment transfer.

Quantitative extraction. From each of the 80 selected papers we extracted reported power-law exponents, saturation thresholds, and data-multiplier effects. We recorded relationships between input factors (parameters, trajectories, embodiments, FLOPs) and output metrics (test loss, success rate, task accuracy), prioritizing primary empirical fits over extrapolated projections.

Synthesis. We synthesized findings along 5 axes: data scaling, model scaling, embodiment diversity, compute scaling in world models, and safety alignment. For each axis we compared exponents and transition points across the corpus, noting deviations from NLP baselines [1, 2]. The synthesis generated a 15,670-word technical report and a conceptual comparison figure.

Limitations. The retrieval depends on arXiv preprints because the Semantic Scholar API was inaccessible, so the corpus may underrepresent peer-reviewed literature. Title-level clustering cannot capture methodological details available only in full text. The absence of parsed abstracts or venue metadata during retrieval prevented automatic quality stratification, and inclusion rested on relevance to quantitative scaling-law measurement rather than on citation impact.

4 Results

We organize the synthesized findings along four axes: data scaling, model capacity, embodiment diversity, and safety.

Data scaling laws are shallower than in language modeling. Power-law relationships between dataset size and test loss hold for robotic manipulation but with shallower exponents. Manipulation loss scales as $L(D) \propto D^{-0.15}$ to $D^{-0.20}$ with training trajectories, whereas language modeling loss scales as $L(D) \propto D^{-0.095}$ [1]. This difference reflects the higher sample complexity induced by continuous action spaces and physical variance. Data diversity dominates raw trajectory count. Cross-embodiment aggregation yields positive transfer, and the Open X-Embodiment dataset demonstrates that pooling heterogeneous robot data improves generalization [3]. Synthetic augmentation of grippers, viewpoints, and backgrounds increases effective dataset size by factors of 2 to 4 [12, 16]. These results indicate that diversity-aware curation scales more efficiently than uniform data collection.

Model scaling exhibits architecture-dependent saturation. Vision-language-action models improve with capacity up to 3 billion to 7 billion parameters and then plateau on current robotic datasets. RT-2 [4] and OpenVLA [5] follow this trend. In contrast, diffusion-based policies scale more favorably for continuous control and show less saturation with increasing data volume [6, 10]. Early generalist models established cross-task viability: Gato handled heterogeneous control tasks with a single transformer [7], while RT-1 [8, 20] and Octo [9, 17] extended this approach to high-capacity vision-language-action architectures. Table 1 summarizes these architecture-specific trends.

Embodiment diversity introduces a new scaling axis. Success rates in locomotion scale with exponents of approximately 0.15 to 0.20 with respect to the number of training embodiments [11, 19]. Cross-embodiment world models require sublinear capacity growth $O(N^{0.5})$ to maintain per-embodiment performance as morphology count N increases. World models such as DreamerV3 [14] reveal a trade-off between imagination depth and data efficiency. Longer prediction horizons and larger latent dimensions improve long-horizon planning but produce steeper compute scaling curves than pure language modeling. Consequently, compute-optimal allocations from NLP [2] do not transfer directly to embodied settings, where environment interaction and simulation dominate cost.

Safety and human-centric data follow distinct scaling curves. Larger cross-embodiment models do not automatically yield safer or more calibrated decisions [15, 19]. Benchmarks show that bigger vision-language models can become brittle under sensory shift, and alignment properties must be scaled explicitly. Human-centric data offers favorable scaling for humanoid morphologies. Human video pre-training is 2 to 4 times more sample-efficient than teleoperated robot data, establishing a strong scaling curve for humanoid control.

Table 1: Scaling trends across embodied learning axes.

Axis	Policy Class	Scaling Trend	Saturation Point	Source
Data (manipulation)	Imitation learning	$L(D) \propto D^{-0.15}$ to $D^{-0.20}$	Not reported	[11, 12]
Data (language)	Transformer LM	$L(D) \propto D^{-0.095}$	Not reported	[1]
Model size	VLA models	Positive up to 3B–7B parameters	3B–7B parameters	[4, 5]
Model size	Diffusion policy	Favorable continuous control scaling	Less saturation	[6, 10]
Embodiment count	Locomotion	$\text{Success} \propto N^{0.15}$ – $N^{0.20}$	Sublinear capacity	[11, 19]

5 Discussion

The evidence synthesized from 80 papers indicates that scaling laws from natural language processing provide a useful but incomplete template for embodied intelligence. Kaplan et al. established that language modeling loss scales as a power law with dataset size with an exponent of approximately -0.095 [1], and Hoffmann et al. showed that model size and training tokens must grow in tandem to avoid compute suboptimality [2]. In robotics, the exponents are markedly shallower. Manipulation loss scales as $L(D) \propto D^{-0.15}$ to $D^{-0.20}$ with trajectories, reflecting the higher sample complexity imposed by continuous action spaces and physical variance [11, 12]. This divergence implies that merely projecting NLP scaling curves onto robot learning overestimates returns from raw data volume. The field must therefore derive embodied-specific scaling constants that account for action-space heterogeneity and environmental stochasticity.

Data diversity emerges as a more influential variable than raw trajectory count. The Open X-Embodiment dataset demonstrates that aggregating heterogeneous robotic data produces positive transfer across morphologies and tasks [3], and cross-domain frameworks such as Mirage confirm that pooling diverse embodiment data improves generalization [13]. Synthetic augmentation, including variations in grippers, camera viewpoints, and scene textures, acts as an effective data multiplier. These findings suggest that data collection strategies should prioritize environmental and object diversity over repetitive demonstration of single-task trajectories. The scaling law for embodied intelligence is therefore not merely a function of dataset cardinality but of the diversity envelope within which the data are sampled.

Model capacity scaling exhibits architecture-dependent saturation. Vision-language-action models such as RT-2 and OpenVLA improve with scale up to 3 billion to 7 billion parameters on current robotic datasets, after which performance plateaus [4, 5]. In contrast, diffusion-based policies for continuous control scale more efficiently, showing reduced saturation as data volume increases [6, 10]. This discrepancy indicates that the choice of policy class determines how effectively additional parameters and data translate into control performance. Generative policy architectures may offer a more scalable substrate for high-frequency continuous action spaces than discrete token-based VLA transformers.

Embodiment diversity constitutes a distinct scaling axis that has no direct analog in language modeling. Empirical studies in locomotion reveal that success rates scale with exponents of 0.15 to 0.20 with respect to the number of training embodiments, and cross-embodiment world models require sublinear capacity growth—on the order of $O(N^{0.5})$ —to maintain per-embodiment performance as the number of morphologies N increases [3, 13]. These relationships confirm that pooling data across robot types yields gains beyond what any single embodiment can achieve in isolation. Future model designs must therefore incorporate mechanisms for conditioning on or routing across heterogeneous embodiment specifications without incurring linear capacity costs.

Compute scaling in world models introduces trade-offs between prediction depth and data efficiency. DreamerV3 shows that extending imagination horizons and expanding latent dimensions improve long-horizon planning, but the cost of observation compression produces a steeper compute scaling curve than that observed in pure language modeling [14]. The need to compress high-dimensional sensory observations into predictive latents adds a multiplicative overhead that training-time FLOPs estimates alone do not capture. Researchers must co-design latent world models with compute budgets, recognizing that deeper imagination carries embodied-specific costs not predicted by NLP scaling frameworks.

Safety and alignment do not scale automatically with model size or data volume. Larger cross-embodiment policies can become brittle under sensory perturbations, and increases in VLA capacity do not inherently produce safer physical decisions. This gap indicates that reliability, calibration,

and risk awareness must be treated as explicit scaling axes rather than assumed byproducts of scale. Future empirical studies should quantify how alignment metrics vary with model size, data diversity, and embodiment count to establish safety-aware scaling laws.

Limitations. This survey faces several methodological constraints. The initial Semantic Scholar API queries returned zero retrievable records due to HTTP 403 errors, forcing reliance on arXiv as the primary source. The final corpus of 80 papers was selected from 99 unique arXiv records and an expanded pool of 333 candidates, with a visible skew toward 2024–2025 preprints. This recency bias limits coverage of peer-reviewed foundational work published in archival venues. Because the retrieval pipeline operated at the title level without systematic parsing of abstracts or venue metadata, the thematic clustering reflects topical word frequency rather than deep semantic analysis. The findings therefore represent current preprint trends and should be updated as peer-reviewed empirical studies on embodied scaling become available.

6 Limitations

Our literature search encountered API failures that shaped the final corpus. Eight queries to the Semantic Scholar API returned HTTP 403 Forbidden and yielded 0 retrievable records. We recovered 99 unique papers from an initial arXiv search and 333 from a second, from which we selected 80. This recovery strategy skews the corpus toward recent preprints posted on arXiv, particularly from 2024 and 2025, and away from peer-reviewed venue publications that may not appear there. Because the retrieval phase collected only titles and did not parse abstracts, author lists, or venue metadata, we could not weight papers by citation impact or peer-review status.

The reported scaling exponents vary in measurability and context. Manipulation loss scales as $L(D) \propto D^{-0.15}$ to $D^{-0.20}$ in some studies [11, 12], whereas language modeling loss scales as $L(D) \propto D^{-0.095}$ [1]. These figures derive from different action spaces, observation modalities, and task distributions, so direct comparison assumes commensurate sample complexity rather than proving it. Few papers report exact compute budgets or data-acquisition costs, which makes cost-aware scaling comparisons impossible.

Vision-language-action models such as RT-2 and OpenVLA improve up to 3 billion to 7 billion parameters before plateauing [4, 5], and diffusion policies appear to scale more favorably for continuous control [6, 10]. Yet these trends depend on the specific datasets and evaluation protocols in each study. No standardized benchmark systematically varies model size, data volume, and embodiment count under identical hardware and task conditions. Consequently, observed phase transitions in model scaling may reflect dataset saturation rather than fundamental capacity limits.

Most scaling studies measure task success rate or imitation loss rather than safety-critical behavior under distribution shift. Larger cross-embodiment models can become brittle under sensory perturbation, and benchmarks show that bigger vision-language models do not automatically make safer embodied decisions [15]. The survey also identifies a shortage of work on federated or modular scaling for heterogeneous fleets and on temporal scaling for long-horizon tasks. These gaps mean our conclusions about scaling laws rest primarily on short-horizon, single-task or pooled-dataset experiments rather than on extended real-world deployments.

7 Conclusion

Scaling laws from natural language processing provide a productive but incomplete template for embodied intelligence. Kaplan et al. established that language modeling loss decays as a power

law with dataset size with an exponent near -0.095 [1], and Hoffmann et al. showed that model parameters and training tokens must grow in equal proportion to reach compute optimality [2]. In robotics, the same functional form holds, yet the empirical exponents are shallower. Manipulation loss scales as $L(D) \propto D^{-0.15}$ to $D^{-0.20}$ with trajectories, a direct consequence of continuous action spaces, embodiment heterogeneity, and physical variance [11, 12]. These findings imply that purely extrapolating NLP training budgets into robot learning overestimates sample efficiency and underestimates the value of data diversity.

Data diversity, rather than raw trajectory count, drives generalization. Aggregating heterogeneous robot data yields positive transfer, and pooling across embodiments effectively multiplies dataset size [3]. On the model side, vision-language-action transformers such as RT-2 and OpenVLA improve with scale up to approximately 3×10^9 to 7×10^9 parameters before performance saturates on current corpora [4, 5]. In contrast, diffusion-based policies for continuous control exhibit more favorable scaling curves and show less saturation as data volume increases [6, 10]. These results suggest that the choice of architectural substrate determines how faithfully model capacity translates into embodied competence.

Embodiment diversity itself constitutes a new scaling axis. Success rates in locomotion scale with exponents of roughly 0.15 to 0.20 with respect to training embodiment count, and cross-embodiment world models require sublinear capacity growth to maintain per-embodiment performance [13]. Generalist agents such as Gato and RT-1 demonstrated that a single transformer can absorb heterogeneous control tasks [7, 20], while open-source successors such as Octo extended these architectures across robot morphologies [17]. World models including DreamerV3 further reveal that longer imagination horizons improve long-horizon planning but steepen the compute scaling curve [14]. Together, these advances mark a shift away from isolated task-specific controllers toward Internet-scale cross-embodiment foundation models.

Several open challenges remain. The field lacks standardized benchmarks that systematically vary model size, data volume, and embodiment count under identical conditions. Cost-aware scaling metrics must account for data acquisition and hardware wear, not merely floating-point operations. Temporal scaling for long-horizon tasks, federated training across heterogeneous fleets, and alignment scaling for safety and calibration require explicit study rather than implicit assumptions. Resolving these questions will determine whether scaling laws in embodied intelligence converge toward the predictability already achieved in language modeling.

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